Credit Risk Portfolio Monitor: solution overview

A Streamlit dashboard over 14,685 loans / 10,000 customers. It has a portfolio view, an account view, a default model and two embedded AI features. Built from 5 synthetic tables, 1 row per account after aggregating bureau, repayment and collections data.

What the dashboard shows and how an analyst uses it

Sidebar filters (loan type, employment, region, income band, age band, risk band, city tier) drive the portfolio view. **Portfolio:** KPIs (exposure, default rate vs portfolio, defaulted exposure, avg PD, flags), risk segments (exposure + actual default by model band), a dimension explorer, a cross-dimension heatmap, a quarterly origination trend, and a segment → account **drill-down**. The analyst uses it to see where losses concentrate and which segments drift. **Account view:** PD, band, top-5 SHAP drivers, 12-month repayment timeline, bureau lines and collections. Used to explain a score before a limit or collections decision. **Ask the Portfolio** and **Anomaly review** (below). **Model & data quality:** metrics, SHAP, calibration by band, monitoring plan and the full DQ log.

AI feature 1: Ask the Portfolio (LLM)

How: the LLM (openai/gpt-oss-120b on Groq's free tier) gets the question plus 4 tools (query_portfolio, list_dimension_values, get_account, get_model_info). It never sees raw tables. The tools compute in pandas with the same functions as the dashboard and return formatted figures. The LLM answers in 1-4 sentences with a *Source* line. A post-check matches every number in the answer to a tool result; misses are badged UNVERIFIED, and the tool trace is shown. Out-of-scope questions (forecasts, profit, names) are declined. **Trigger:** analyst submits a question. **False positive:** a confident answer built on the wrong filter or metric (visible in the trace). **False negative:** an answerable question declined, e.g. an odd label spelling. **Human step:** analyst checks the Source/trace before reusing a figure. **No API key:** LLM disabled; a manual query builder runs the same engine.

AI feature 2: Anomaly detection

Trigger: any of (a) *own history*, months 10-12 vs 1-9: DPD jump to ≥30 from ≤5, payment ratio drop from ≥95% to <50%, 3+ consecutive misses; (b) *peer segment* (loan type × income band): EMI-to-income robust z > 3.5, bureau 30+DPD / written-off / new lines at or above the peer 99th percentile, utilization ≥90% where the peer median is <50%; (c) Isolation Forest top 2%. **1,207 flagged**, ranked by severity, each with plain-English reasons. Flagged default rate is 22.8% vs 15.5% unflagged. **False positive:** a benign one-off (bank-side payment failure; a business loan serviced from revenue). **False negative:** slow deterioration in line with peers, or stress outside our data. **Human step:** nothing is auto-actioned. The analyst Confirms / Dismisses / Escalates with a note, logged with a timestamp to tune thresholds.

Model results and approach

Model (30% hold-out, n=4,406)	AUC	KS	Gini
Logistic regression (baseline)	0.6611	0.2379	0.3221
LightGBM (chosen)	0.6711	0.2457	0.3422
LightGBM 5-fold out-of-fold (all accounts)	0.6785	0.2625	0.357

40 features: application, engineered bureau (write-offs, DPD, utilization, new lines, thin-file) and repayment (DPD level and trend, pay ratio, misses, late count). Collections are excluded as leakage. Scoring point = month 12. LightGBM is shallow and CV-tuned (deeper trees overfit: train 0.89 vs test 0.66). The signal is largely monotonic, so LR is close behind. Top SHAP features: income, years employed, repayment DPD, bureau DPD, age. Bands are well calibrated (actual default rises Low→Very High). **Monitoring:** monthly score/feature PSI (alert 0.10 / act 0.25), Gini on matured cohorts (alert at -5 pts), calibration by band, DQ checks, fairness by age/gender.

Top 3 portfolio insights

Mortgages are the loss concentration. Mortgages default at 24.7% (portfolio 16.1%). They hold 52% of exposure but 65% of defaulted exposure. *So what:* Mortgage underwriting is the biggest lever on losses. Review LTV/affordability cut-offs and put mortgages first in the early-warning queue, even though they are secured.

Income is the strongest single driver. Default rate falls from 38.5% under $30k to 7.9% above $120k, a 4.9x gap. It is also the top SHAP feature in the model. *So what:* Use income-based affordability limits (loan-to-income, EMI-to-income) in approval, and verify income for applicants under $60k.

Cross-cut: employment type changes risk inside the same income band. In the $30-60k band, Contract workers default at 26.8% and Unemployed at 33.1%, vs Salaried 21.1%. Unemployed overall: 35.0% on 718 accounts. *So what:* Price and set limits on income x employment together, not income alone. Unemployed applicants need a documented alternative repayment source.

Key assumptions and what production needs from the bank's IT and data teams

Assumptions: target_default is a fixed historical outcome observed after month 12. Collections dates that precede origination and customer_since dates after origination are data errors (flagged and kept, not used as features). Bureau DPD nulls mean unknown, not zero.
Needed for production: governed source feeds (core banking loans/repayments daily, bureau pull API, collections system) with keys and timestamps; a point-in-time feature store to avoid leakage; a scheduled pipeline (Airflow/dbt) and model registry; a database table for review decisions in place of the CSV; SSO + role-based access; an approved LLM endpoint (private / VPC, zero data retention) with logging; model risk management validation; data-privacy sign-off on which aggregates may leave the bank.